

Data Science 6

Mark

Lecture 5

Monday, June 30, 2026

Day 5

You can populate a table with `with_columns` using the name and array. Note a objects function vs an attribute.

You can extract rows from a table by

1. `tbl.where()` by condition
2. `tbl.take()` by index

for `tbl.take()` use `np.arange()` ex: `table.take(np.arange(5))`. `np.arange` works by returning an array usual start step but stop is not included in the array.

Note. Must have all positional arguments or all keyword arguments

1 Booleans

boolean predicates use the "are." Usually can just use lambda functions Recall the comparison operators from earlier

Comparison Operators

Operator	Meaning
<code>==</code>	Equal to
<code>!=</code>	Not equal to
<code>></code>	Greater than
<code><</code>	Less than
<code>>=</code>	Greater than or equal to
<code><=</code>	Less than or equal to

Figure 1: Python Ops

Falsey Values:

1. False
2. 0

3. 0.0
4. "" (empty string)
5. None

Everything else is Truthy. `int(True) = 1`

Note. Dont compare floates using `==` because of floating point error. and use instead `abs(0.1 * 6 - 0.6) < 0.0001`

2 Functions

Recall how to make a function. can use a doc string if feeling like you have time to waste

Recall string methods

Python String Methods

Input	Method	Output
'hello WORLD'	.capitalize()	Hello world
'HELLO WORLD'	.lower()	hello world
'hello world'	.upper()	HELLO WORLD
'Python'	.center(10, '*')	**Python**
'HELLO WORLD'	.count('L')	3
'HELLO WORLD'	.index('O')	4
'HELLO WORLD'	.find('OR')	7
'31/01/2022'	.replace('/', '-')	31-01-2022
'31/01/2022'	.split('/')	['31', '01', '2022']
'abc123'	.isalnum()	True
'12345'	.isnumeric()	True
'hello world'	.islower()	True
'HELLO WORLD'	.isupper()	True

Figure 2: String Methods